

Some popular embedded RTOSes

LinuxOS. A Unix-like real-time operating system from LinuxWorks. It features full POSIX conformance and Linux compatibility

QNX Neutrino. Supports ARM, MIPS, PowerPC, SH and X86 processor families

VxWorks. A popular RTOS offered by Wind River, an Intel group company. Supports a range of architectures like ARM, PowerPC, ColdFire and MIPS

INTEGRITY. A hard RTOS from Green Hills Software Inc, that supports ARM, PowerPC, ColdFire, x86, and so on

ThreadX. Express Logic's advanced RTOS designed specifically for deeply embedded applications. It supports ARM, PowerPC, x86, etc

Windows CE. Microsoft's offering for embedded systems and minimalist computers, this is supported on Intel x86 and compatibles, MIPS, ARM, and Hitachi SuperH processors

Nucleus RTOS. A product of Mentor Graphics, Nucleus RTOS is a highly-modular, hardware-independent RTOS that has been ported to a number of ARM, Power, and MIPS architecture central processing units

Red Hat Embedded Linux. An open source offering based on the Linux kernel, this has been used for a wide range of purposes, including navigation systems

eCos. An open source RTOS that supports ARM, Hitachi H8300, Intel x86, MIPS, Matsushita AM3x, Motorola 68k, PowerPC, SuperH, SPARC, NEC V8xx, etc

FreeRTOS. FreeRTOS.org is a portable, open source, mini real-time kernel that supports more than a dozen architectures including ARM, AVR32 and MSP430

Symbian OS. A leading open operating system for advanced data-enabled mobile phones. Suits advanced 3G mobile phones also

Stable Hybrid Release. SHR, available in full and light versions, is a Linux-based RTOS developed and tested for ARM-based OpenMoko phones, although it can be ported to other architectures as well

Android. A mobile operating system based on the Linux kernel. Google changed the way the RTOS game was being played, with this successful release. Many consider it a market disruption. Several brands have launched Android phones now

such high-performance metrics and exploit the functionality of the silicon. But Wind River believes that RTOS requires at least 32-bit microprocessors, and we focus on this segment and higher. The most significant feature of our most recent release, VxWorks 6.7, is 64-bit computing," says Kumaran.

64-bit RTOSes usually support 32/64-bit embedded microcontroller units, such as ARM, MIPS, Power Architecture, 68K/Coldfire, x86, SuperH, etc. The company is also capable of working on such platforms.

Android in the RTOS space

"Any discussion around the biggest market disruptions that we have seen in the recent past would have to begin with the mention of Android. It is a game-changing initiative that also saw Wind River's serious foray into the mobile market," says Kumaran. "We have also seen embedded platforms moving to the enterprise computing segment, and the acceptance of Linux.

Wind River is definitely gaining traction in this space as the market moves to advanced telecommunications computing architecture/high availability (ATCA/HA) platforms. From the defence perspective, a monumental change has been observed in certifying platforms on a VxWorks 653 operating environment."

Coming back to Android, it is a Linux-based open source operating system for mobile phones. It was launched by Google and then taken over by the Open Handset Alliance. There are now more than a dozen mobile phone models, including some by Samsung, LG and HTC, that use Android. As a result, it has now proved its worth, and is gaining traction. RTOS majors, including Wind River and Nucleus (Mentor Graphics), are now offering Android-based products including development tools.

With Linux and Android in the picture, open source is becoming more popular in the RTOS space.

IPv6 support and other advancements

Basically, embedded RTOSes enable developers and manufacturers to make the best use of the microprocessor board to make their device perform to the optimum level. Hence, if a new feature is required in a product, not only the microprocessor unit but also the embedded RTOS should be able to support it. In that sense, RTOSes are constantly evolving to support new standards and features, ranging from USB 3.0 to Internet Protocol version 6.0 (IPv6).

The Nucleus RTOS, for example, recently rolled out support for IPv6, including certification under the IPv6 Ready Logo (Phase 2) program. It means that the developers who need to integrate their products on IPv6 networks have a certified protocol stack that can be interoperable with other IPv6 networked devices.

More in store

Although hidden somewhere within modern devices, the world of embedded systems is growing, providing an impetus to the growth of the operating system that powers it.

For an example, we have seen that how 64-bit microprocessors that were once used in supercomputers and entered the desktop computing space around 2003, have already become a part of embedded systems. The RTOSes have also quickly adapted to make the most of these powerful systems. Another example of how fast the RTOS world reacts to market changes is its reaction to the success of Android. Almost every market leader has now accepted Android within its folds and offers products for, or based on it.

Therefore, although silent, the embedded RTOS space has shown and continues to show significant growth and innovation, and is worth a frequent look-up. ●

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